

Trial extraction — how the modules interact

Deterministic code owns the control flow; the LLM fills two kinds of slot — a **doer** and a **reviewer** — on a bounded refine loop. Everything runs once per trial, streamed to one output.

GUIDING PRINCIPLE — ONE AGENT PER INDEPENDENT TASK

Split work into a separate agent only when the sub-tasks are **genuinely independent**. Outputs that must be decided **together** stay in one agent (a DNF — disjunctive normal form — row is a conjunction across columns → **one extractor emits whole rows**). Independent checks **fan out** (→ a **panel of focused reviewers**). Deterministic work gets **zero** agents.

■ Deterministic code
 ■ LLM · doer (extract / map / research)
 ■ LLM · reviewer

SELECT & ASSEMBLE Load trial(s); build one labelled document

[CODE](#)

ID= (source auto-detected), IDS= (a set), or none = all. Sections become **## LABEL** headers — each becomes a cell's `[source]` tag.

COHORTS & DRUG Split by registry — sources differ

[CODE + LLM](#)

CTGOV · DETERMINISTIC CODE

one cohort per `armGroups` entry: label, `arm_type`, drug from `interventionNames`. No LLM.

ANZCTR · 2 LLM DOERS

cohort-detection (conservative; default single) + **drug** extraction from the `INTERVENTIONS` text.

STAGE I — EXTRACTION extract → check → refine · the bounded evaluator–optimizer loop at its heart

One agent **generates** the candidate table; a panel of agents **verifies** it. Verification is an easier task than generation and runs on a fresh pass with the finished table in view — so it catches omissions the generator missed. On a gating failure the loop does **incremental repair**: the extractor gets its OWN previous table + only the flagged issues and **keeps unflagged rows verbatim** (preserves correct work, aids convergence), up to 3 attempts.

DOER · EXTRACT

Reads the assembled document + cohort list → the candidate DNF table (whole rows, so cross-column co-occurrence is preserved).

ESSENTIAL TASKS

- > one row = one ANDed conjunction; OR-alternatives → separate rows
- > fill 5 columns (cancer_type, gene_alteration, molecular_signature, molecular_biomarker, prior_therapy); "" when not required
- > exclusions inline as NOT(...) ; same-column carve-outs stay in one cell
- > never write a self-contradiction X AND NOT(X) in one cell — if the same thing is required for one cohort and excluded for another, **split into rows**
- > cancer_type: CONDITIONS is authoritative; **never AND two different tumour types**; drop a broad umbrella when the trial is clearly one specific type
- > tag every value with its source section(s) [a; b]
- > assign each row a cohort scope: a cohort id or trial-wide

EXAMPLES

```
[trial-wide] cancer_type = "ES-SCLC AND NOT(complex SCLC) [TITLE; ELIGIBILITY CRITERIA]"
```

```
[C1] molecular_biomarker = "HR+ AND HER2- [ELIGIBILITY CRITERIA]"
```

```
[trial-wide] prior_therapy = "NOT(prior anti-PD-1) [EXCLUSION CRITERIA]"
```

▼ candidate table

code check deterministic rule-check — at least one row, and no row with all five eligibility columns empty; a failure goes straight back to the doer (no reviewer call).

REVIEWER · 5-AGENT PANEL (PARALLEL)

Each reviewer sees the whole candidate table + source and judges ONE dimension → faithful? + problems. Independent checks → separate agents (the guiding principle).

ONE REVIEWER PER DIMENSION

cancer_type	tumour under study ? reject one seen only in a prior-therapy / history / exclusion line (e.g. drop "melanoma" from a lung trial); flag ANDing two different types or an un-dropped umbrella. gates
molecular	faithful + right column (HER2 IHC → biomarker, ERBB2 amp → gene, MSI-H → signature); flag a cell's X AND NOT(X) (conflated cohorts → split rows). gates
prior_therapy	required / excluded only (omit merely-permitted); exclusions wrapped in NOT() . gates
structural	DNF integrity: conjunctions / OR-rows correct; each row's cohort scope right; no X AND NOT(X) cell. gates
drug	each cohort's drug reflects its arm's intervention. advisory (non-gating)

Refine (incremental) — any of the 4 gating reviewers fails → **Extract** re-runs with its OWN prior table
🔄 + only the flagged issues, keeping unflagged rows as-is (the drug reviewer is advisory, so it never blocks). Bounded at 3 attempts.

✓ PASS

rules pass **and** all 4 gating reviewers faithful → leave the loop.

⚠ REDO

a gating reviewer fails → re-extract with its feedback, re-check.

■ STOP

after 3 attempts → exit as **faithful=False**, best effort, written & logged.

STAGE II – MAPPING

on the extracted rows · each procedure = doer → code check → reviewer (independent task)

Each normalized cell is mapped to a controlled representation, then reviewed. Curated resources ground the prompts + tests but are held out as verification data (not ingested) — so it stays a mapping, not a vlookup.

cancer_type → OncoTree

- doer** expression → OncoTree name + code (e.g. "metastatic NSCLC" → NSCLC). Only 3 non-OncoTree terms allowed: Pan-cancer / solid tumour / Haematological malignancy (no [None]).
- code check** reject invalid codes and logic errors: $X \text{ AND } X$, $X \text{ AND NOT}(X)$, a broad term ANDed with its subtype, a subtype ANDed with its parent (hierarchy via `is_subcode`).
- reviewer** right node + granularity; structure preserved; NOT(cancer type) kept minimal.

grounds: OncoTree ontology + hierarchy (tools/oncotree.py) + a few ConditionsCurationResource examples

gene_alteration / molecular_signature → finding-model

- doer** → finding-model syntax: `SmallVariant[gene=BRAF & ...=p.V600E]` , `Fusion[geneEnd=ALK]` , `MicrosatelliteStability[...=MSI]` .
- code check** balanced brackets, known classes, gene-scoped SmallVariant; no duplicate terms, no $X \text{ \& NOT}(X)$ self-contradiction.
- reviewer** faithful + valid grammar (strict); a NOT() qualified by something inexpressible (e.g. a location) is **omitted**, not negated.

grounds: finding-model grammar (tools/finding_model.py) + GeneAlteration/MolecularSignature examples

drug → main/aux + class + TGA/PBS

- doer (web)** main = the **investigational agent(s) by judgement** (not the whole regimen; backbone/SoC/comparator/placebo → auxiliary); POTTR + general class; **per main drug** TGA & PBS status (Approved/Not approved) + a detail column (year + evidence + official link), via **web search**.
- reviewer** main = investigational only; class sensible; per-drug status + detail present.

grounds: POTTR dataset + live web research (Responses API web_search)

WRITE Stream one enriched TSV

[CODE](#)

21 columns (incl. per-drug `tga_status` / `pbs_status` + `tga_detail` / `pbs_detail`), one row per satisfiable (trial, cohort) conjunction; rows appended as each trial completes. A trial that errors is logged & **skipped** — the batch continues, and the summary reports `N ok, M failed` .

SUPPORTING LAYERS · CORE/ + TOOLS/

`client.py`

One door to OpenAI: `.parse()` (structured) + `.research()` (web_search); cache, retries, tracing.

`agent.py`

Agent = prompt + schema + model (+ web_search flag), bound to the client.

`workflow.py`

`fan_out` (parallel) + `refine` (bounded check→incremental-repair loop).

`logfmt.py`

Shared run-log formatting: stage banners + doer/reviewer blocks (one field per line, spelled-out PASS/FAIL).

`tools/`

`oncotree.py` (vocab + validator + hierarchy/is_subcode, 3 sentinels) · `finding_model.py` (grammar + syntax/logic validator).